Scalability & Future Plan

Date	15 July 2026
Team ID	SWTID-2026-4735
Project Name	Credit card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Supports a limited number of users on a single Flask server.	Deploy using Gunicorn with multiple workers behind Nginx or a cloud load balancer to handle more concurrent users.
2	**Data Storage**	Uses a trained `.pkl` model and does not store user prediction history.	Integrate a database such as MySQL or PostgreSQL to store applicant details and prediction records.

3	**Performance**	Predictions are generated in real time using the Random Forest model on a single server.	Optimize the model, enable caching, and deploy on scalable cloud infrastructure with higher compute resources.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Model is trained on a limited historical dataset and may not generalize well to unseen real-world data.	May reduce prediction accuracy for applicants with different data patterns.	**High**
2	The application does not integrate with real-time credit bureau or banking APIs.	Predictions are based only on the trained dataset and not on current financial records.	**High**
3	The system provides only the approval/rejection result without explaining the reason behind the prediction.	Users and financial institutions cannot easily interpret the model's decision.	**Medium**

Scalability Plan:

4	**Security**	Basic form validation is implemented; no user authentication or data encryption.	Add HTTPS, user authentication, input sanitization, and encrypt sensitive data for secure access.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate a database (MySQL/PostgreSQL) to store applicant details and prediction history.	1–2 Weeks	Enables persistent data storage, reporting, and record management.
Phase 3	Add Explainable AI (XAI) and integrate real-time credit bureau/banking APIs.	2–4 Weeks	Improves prediction transparency and increases decision accuracy using live financial data.
Phase 4	Deploy on a scalable cloud platform with user authentication and an admin dashboard.	4–6 Weeks	Enhances security, supports multiple users, and makes the application production-ready.